

COMP2040 – Operating Systems

C Environment Setup: macOS, Linux, Windows

You need a terminal, the `gcc` compiler, and an editor (we recommend Sublime Text). The labs use POSIX system calls (`open`, `read`, `stat`, ...), which Windows lacks natively, so Windows users need WSL or Cygwin. Do your section, then Section 4 and Section 5.

1 macOS

```
| xcode-select --install  
| gcc --version
```

The output says *Apple clang*: on macOS, `gcc` is Clang. This is fine for the course.

2 Linux

Ubuntu/Debian	<code>sudo apt update && sudo apt install build-essential gdb manpages-dev</code>
Fedora	<code>sudo dnf install gcc make gdb man-pages</code>
Arch	<code>sudo pacman -S -needed base-devel gdb man-pages</code>

Then check with `gcc -version`.

3 Windows

Choose **one** option. WSL is closest to a real Linux system; Cygwin is simpler to install and needs no virtualisation or restart.

3.1 Option A: WSL (Ubuntu)

Requires Windows 10 (2004 or later) or Windows 11.

1. Open PowerShell as Administrator and run `wsl -install`.
2. Restart the computer.
3. Open Ubuntu from the Start menu and create a Linux user name and password.
4. In Ubuntu, install the compiler:

```
| sudo apt update && sudo apt install build-essential gdb manpages-dev  
| gcc --version
```

If step 1 or 3 fails with a virtualisation error (e.g. `0x80370102`), either enable virtualisation (*Intel VT-x / AMD SVM*) in the BIOS, or switch to WSL 1, which does not need it:

```
| wsl --set-default-version 1  
| wsl --install -d Ubuntu
```

If it still fails, use Option B.

Where to work. Keep lab files in the Linux home folder (`mkdir ~/comp2040`), not under `/mnt/c`. On `/mnt/c`, `chmod` does not work properly and Task 5 of Lab 01 will give wrong results. To open the folder in Windows, run `explorer.exe` . from it.

3.2 Option B: Cygwin

1. Download `setup-x86_64.exe` from <https://cygwin.com>.
2. Run it and accept the defaults (Install from Internet, root `C:\cygwin64`), then pick any download site.
3. On Select Packages, set View to Full, search for each package below, and change Skip to the latest version: `gcc-core`, `make`, `gdb`, `man-db`.
4. Finish the installer and open Cygwin64 Terminal:

```
gcc --version
mkdir ~/comp2040
```

Where to work. Your Cygwin home folder is `C:\cygwin64\home\username`. Keep lab files there and open them in Sublime Text directly.

Differences from Linux.

- `gcc ... -o task1_hello` creates `task1_hello.exe`; run it as `./task1_hello` as usual.
- Permissions are mapped onto Windows ACLs. `chmod` works, but group/other bits may differ from Linux. Always compare your program's output with `ls -l`.

4 Sublime Text

Install.

- macOS, Windows: download from <https://www.sublimetext.com/download>.
- Linux: follow https://www.sublimetext.com/docs/linux_repositories.html.

Settings. Open Preferences > Settings and put this in the right-hand (user) pane:

```
{
  "tab_size": 4,
  "translate_tabs_to_spaces": true,
  "ensure_newline_at_eof_on_save": true,
  "default_line_ending": "unix"
}
```

The last line matters on Windows: it prevents Windows line endings in your C files.

Compile from the terminal. You may set up a Sublime build system, but the labs expect you to compile and run in the terminal.

5 Check your setup

Save this as `check.c`:

```
#include <stdio.h>
#include <unistd.h>
```

```
#include <sys/stat.h>

int main(void)
{
    struct stat st;

    if (stat("check.c", &st) == -1) {
        perror("stat");
        return 1;
    }
    write(STDOUT_FILENO, "system calls work\n", 18);
    printf("check.c is %lld bytes\n", (long long) st.st_size);
    return 0;
}
```

Then, in the same folder:

```
gcc -Wall -Wextra check.c -o check
./check
```

You should see `system calls work` and the file size, with no warnings.

6 Common problems

Problem	Fix
gcc: command not found	Compiler not installed, or you are in PowerShell instead of the Ubuntu/Cygwin terminal.
macOS: xcrun: error: invalid active developer path	Run <code>xcode-select -install</code> again (common after macOS updates).
unistd.h: No such file or directory	You are using a native Windows compiler (MinGW, Dev-C++). Use WSL or Cygwin.
WSL: Permission denied, or chmod has no effect	Your files are under <code>/mnt/c</code> . Move them to <code>~/comp2040</code> .
\r in compiler errors	Windows line endings. In Sublime: View > Line Endings > Unix, then save.

Still stuck? Bring your laptop to the lab or post the exact error on the course forum.